Towards Biologically Inspired Decentralized Platooning for Autonomous Vehicles

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***Abstract*—In this paper, we introduce a novel decentralized approach to automated vehicle platooning inspired by `emergent-behavior` commonly found in biological systems. The proposed emergent-behavior based platooning uses simple rules that the vehicles follow when they interact to carry out maneuvers. Platooning maneuvers such as join, exit, and lane change "emerge" as a result of vehicles following these "emergent rules". Preliminary complexity analysis as well as simulation experiments yield promising results that demonstrate emergent behavior platooning as a viable form of decentralized platooning.**

***Index Terms*—autonomous platooning, decentralized, bio-inspired, emergent behavior**

I. Introduction and Background

According to the U.S. Department of Transportation (USDOT), *platooning* is a coordinated operation of two or more vehicles via cooperative adaptive cruise control (CACC) [1]. Platooning provides a number of benefits, including: fuel efficiency [2], road capacity [3], [4], and road safety. Early research on platooning and automated vehicular systems such as PATH [5], SARTRE [6] and Energy-ITS [7], mainly focused on steady-state cruising stage, which regulates speed and inter-vehicular spacing. The problems of platoon maneuvering such as formation and dissolution have been less explored and only recently have started to receive attention from researchers and practitioners [8].

We introduce a novel decentralized platooning scheme based on a biologically-inspired emergent behavior approach, which uses simple rules that vehicles follow to carry out platooning maneuvers. Maneuvers such as join, exit, and lane change "emerge" as a result of vehicles following these "emergent rules". Additionally, we conduct preliminary complexity analysis as well as simulation experiments that show that emergent behavior based platooning is a scalable and efficient form of decentralized platooning.

II. Autonomous Platooning

In this paper, we focus on autonomous platooning on the freeways where maneuvers can happen during the trip. Platooning in urban centers would require smart road infrastructure, e.g., traffic lights, pedestrian crosswalks that can interact with the vehicles.

A. Types of Platooning Systems

a) **Centralized Platooning:** In centralized platooning systems, all platooning activities are coordinated by a *leader*, usually the vehicle at the front of the platoon. Like any centralized system, centralized platooning suffers from drawbacks such as single point of failure and performance bottleneck (at the leader), as well as shorter platoon-length and data loss due to longer range communication from the leader. A notable example of centralized platooning is described in [9] which we use as a performance baseline in our experiments. Other examples include [10], [11].

b) **Decentralized Platooning:** Decentralized platooning systems were proposed to mitigate the drawbacks of centralized platooning mentioned above. There is no platoon leader and vehicles interact only with neighboring vehicles to complete the maneuvers. Localized interactions allow decentralized platooning systems to handle longer platoons. Examples of decentralized platooning approaches include [8], [12], [13].

c) **Discussion:** Generally, platooning systems adopt top-down approaches where high-level objectives, in this case, the maneuvers, are defined and then workflows, including message exchange, specific for each maneuver are defined. This kind of approach is termed as *Deliberate Systems* [14]. Alternatively, we propose a novel decentralized platooning approach that draws inspiration from nature and adopts a bottom-up approach. The proposed *Emergent System*, also known as *Biologically-Inspired System* [15] first lays out basic rules of interaction, which vehicles would then follow to carry out the platooning maneuvers.

III. Biologically-Inspired Platooning

In nature, organisms are able to build complex structures (e.g., termites), avoid predators (e.g., ants), or move in unison (e.g., fish, starlings), by communicating indirectly using chemical or physical processes, called *stigmergy* [16]. A global behavior that emerges only from local interaction using simple rules with none of the individual organisms having complete view of the end goal is called '*Emergent Behavior*'. The work reported in [17] pioneered the use of emergent behavior inspired by flocking birds to guide robot movement. In the context of vehicle platooning, notable related work includes the system described in [18], [19] which proposes Hierarchical Emergent Behavior (HEB) to tackle large platoons navigating on "unmarked free ground" where vehicles have higher degrees of freedom and therefore cannot be adopted on roads.

A. *Emergent Platooning*

For our emergent-based platooning, we draw inspiration from termites. They communicate indirectly by dropping a chemical called *pheromone* present in their saliva. In an emergent platooning system, vehicles periodically broadcast messages with their current state (including their position, ID, platoon ID, vehicle-role), which are equivalent to pheromones in termites. We call these messages as *gap-pheromones*. While all vehicles hear gap-pheromones from their neighboring vehicles (a la stigmergy), in our current emergent platooning implementation, each vehicle keeps track of gap-pheromone messages sent by its predecessor and successor. Vehicles use: (1) position information of their predecessor to adjust their own speed and acceleration, and when to join the platoon, (2) position information from their immediate successor to decide whether to convert to a tail member.

B. *Why Emergent Platooning ?*

In order to make the case for emergent platooning, we discuss and compare how maneuvers are executed in centralized, "deliberate" decentralized, and emergent platooning approaches. We focus on the **Join** and **Exit** maneuvers, while **Split**, **Merge** and **Lane Change** maneuvers will be considered in our future work.

a) **Join Maneuver: Join** maneuvers enable a non-platoon vehicle to join the platoon. A vehicle that wishes to enter the platoon can join at the head, tail, or anywhere in the middle [11]. In this paper we consider only the tail-join strategy and others will be considered in our future work. In centralized platooning, all maneuvers, including **Joins**, are serialized as they are coordinated by the leader. The total time the leader stays in maneuvering state is $O(n)$, where n is the number of vehicles joining the platoon. In decentralized platooning (both deliberate and emergent), the **Join** maneuver is coordinated by the tail member of the platoon, assuming vehicles join

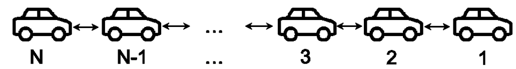


Fig. 1: 1-Predecessor 1-Successor Information Flow

at the end of the platoon. Therefore, each vehicle will coordinate joining of only one other vehicle and total time a vehicle stays in the maneuvering state is $O(1)$.

b) **Exit Maneuver: Exit** maneuvers can be initiated either by the vehicle that intends to leave the platoon or by the vehicle managing the platoon [10]. Any platoon member can exit the platoon. When a middle vehicle exits the platoon, the follower catch-up to maintain constant inter-vehicle gap. Let us consider the case where n vehicles want to exit the platoon simultaneously. In centralized platooning, the maneuver is serialized and the total time for all the n vehicles to exit will be $O(n)$. In "deliberate" decentralized platooning systems, since three vehicles (exiting vehicle, predecessor and successor) are involved in the **Exit** maneuver, $\frac{n}{3}$ vehicles can exit simultaneously, assuming exits happen at different places in the platoon. This means, the total time for all the n vehicles to exit will be $O(6log\ n)$. In the case of emergent systems, the exiter need not communicate with any of the vehicles in the platoon and therefore the total time for n vehicles to exit would be $O(1)$.

IV. Emergent Platooning: System Design

In this section, we describe in detail the design of our emergent platooning system including its controller, maneuvers, and communication protocols.

A. *Controller*

The controller in the vehicle determines the rate of acceleration/deceleration of the vehicle using speed, distance and acceleration of other vehicles that it senses on its own (Adaptive Cruise Control) or by receiving them explicitly via wireless communication (CACC) to maintain desired inter-vehicle spacing. In autonomous platooning, there are two types of inter-vehicle spacing policies used - *Constant Distance Spacing* (CDS) and *Constant Time Spacing* (CTS). CDS requires all vehicles to use speed and acceleration information from the lead vehicle of the platoon to maintain string stability [20]. This restriction does not apply to the CTS policy and the spacing depends on the velocity of the controlled vehicle. While CDS can achieve shorter inter-vehicle spacing compared to CTS, it cannot be used in decentralized platooning [20] since it is inherently centralized as it needs information from the lead vehicle. Therefore, for our decentralized, emergent platooning approach, we use the controller proposed by Ploeg, et al, [21], which

TABLE I: Controller Parameters

Parameter	Description
k_p, k_d, k_{dd}	Constants
$e_{1,i}, e_{2,i}, e_{3,i}$	Space, Speed, Acceleration Errors
V_i	Vehicle i
u_i	Acceleration of V_i
$d_{r,i}$	Inter-vehicle spacing
d_i	Gap between V_i, V_{i-1}
s_i, v_i, a_i, L_i	Position, Speed, Acc, Length of V_i
r	Standstill distance between vehicles
h	Desired time headway
τ	Engine dynamics time constant

is shown by the authors to possess both internal- and string stability properties. This controller is referred to as *ploeg-controller* in this paper. Information flow in our system is as shown in Figure 1, where vehicle$_i$'s predecessor and successor are vehicle$_{i-1}$ and vehicle$_{i+1}$, respectively. Every vehicle i determines its desired acceleration using information from its predecessor according to Equation 1, where controller parameters are summarized in Table I.

$$\dot{u}_i = \frac{1}{h}\left(-u_i + k_p e_{1,i} + k_d e_{2,i} + k_{dd} e_{3,i} + u_{i-1}\right) \quad (1)$$

where,

$$\begin{aligned} e_{1,i} &= d_i - d_{r,i} = (s_{i-1} - s_i - L_i) - (r + hv_i) \\ e_{2,i} &= v_{i-1} - v_i - ha_i \\ e_{3,i} &= a_{i-1} - a_i - h\dot{a}_i \\ \dot{a}_i &= \frac{1}{\tau}(-a_i + u_i) \end{aligned} \quad (2)$$

We assume that all vehicles implement the same controller and therefore, r and h are independent of i. For any value $h > 0$, the errors in Equation 2 can be stabilized with values of $k_p, k_d > 0$ and $k_{dd} > -1$ such that $(1 + k_{dd})k_d > k_p$ [21].

B. Emergent Communication Protocols

There are two types of messages that the vehicles exchange when executing platoon maneuvers - 1. *Gap-Pheromone Messages* and 2. *Maneuver Messages*.

a) **Gap-Pheromone Messages:** These messages are broadcasted periodically by every vehicle in the platoon. The exchange of gap-pheromone messages is illustrated in Figure 1. The time between gap-pheromone broadcasts, or the *epoch* (δt) is a configurable system parameter. Each vehicle keeps a record of gap-pheromone messages sent by its predecessor and its successor. Gap-pheromone messages carry vehicle-id, platoon-id, vehicle-role, in addition to the current position of the transmitting vehicle (v_t) that the receiving vehicle (v_r) uses to compute the gap (g) and the speed (s_t) of v_t. Let $(x_t^0, y_t^0), (x_t^1, y_t^1)$ be the positions of v_t at times (t_0, t_1) and (x_r^1, y_r^1) be the position of v_r at time t_1. At time t_1, $g = \sqrt{(x_t^1 - x_r^1)^2}$ and $s_t = \sqrt{(x_t^1 - x_t^0)^2}/\delta t$. The *gap-pheromone* PP_{gap} for the predecessor and SP_{gap} for the successor are calculated as $1/g$. The vehicle's speed is adjusted based on the speed s_t of its predecessor.

In nature, the pheromone concentration increases as pheromones are deposited and decrease due to evaporation. Taking inspiration from pheromones in nature, at every epoch, if a pheromone message is received from its predecessor, the predecessor's pheromone concentration PAP_{gap} is updated as follows: $PAP_{gap} = (PAP_{gap} + PP_{gap})/2$. Let the threshold ($TPP^L_{gap}$) be equal to the reciprocal of the desired inter-vehicle gap. While PAP_{gap} is less than TPP^L_{gap}, the vehicle will travel at the maximum speed limit, which will be higher than the platoon speed. Once PAP_{gap} is equal to TPP^L_{gap}, the vehicle adjusts its speed to the platoon speed. From this point on, PAP_{gap} will not change and therefore, the vehicle will neither accelerate nor decelerate. When a vehicle stops receiving pheromone messages (because its predecessor exited the platoon), PAP_{gap} decreases exponentially and when it nears zero, the vehicle accelerates to catch-up with its new predecessor.

Similarly, pheromone messages from a vehicle's successor causes the current value of the successor's pheromone concentration SAP_{gap} to be updated as follows: $SAP_{gap} = (SAP_{gap} + SP_{gap}) \times 0.95$. SAP_{gap} is capped at a threshold (TSP_{gap}), equal to the reciprocal of the desired inter-vehicle gap. When SAP_{gap} nears zero, the vehicle assumes that there is no successor and becomes the platoon's tail-member. The fading rate of SAP_{gap} is considerably slower than that of PAP_{gap} such that when a vehicle exits, its predecessor will not prematurely assume the role of the tail member, allowing enough time for the exiter's follower to catch-up.

b) **Maneuver Messages:** Currently, maneuver message supporting **Join** maneuver is implemented in our emergent platooning system. The other maneuvers mentioned in Section III-B will be considered in our future work. Maneuver messages exchanged during a **Join** maneuver and their workflow are shown in Figure 2. Message loss during **Joins** is handled by aborting the maneuver and restarting it, if the state transitions depicted in Figures (2, 4) do not happen within a threshold time. Note that the **Exit** maneuver does not require any maneuvering messages. A vehicle will determine the absence of its predecessor when pheromone concentration drops as a result of not receiving regular pheromone messages from its predecessor.

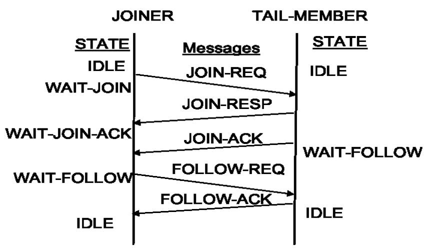


Fig. 2: Emergent Join Maneuver

C. Emergent Platooning Maneuvers

Before describing the **Join** and **Exit** maneuvers, we start by laying out the basic emergent rules used in these maneuvers. When a certain subset of these emergent rules is followed by each vehicle, the corresponding maneuver "emerges", which is what characterizes emergent behavior based systems. Each vehicle is considered to be in one of the following roles - `NON-MEMBER`: a vehicle that has not yet joined the platoon, `JOINER`: a vehicle that is in the process of joining the platoon, `TAIL-MEMBER`: the last vehicle of the platoon, and `IN-MEMBER`: all other vehicles in the platoon.

a) **Emergent Rules:** Any emergent system consists of rules that are implemented by all participants, in our case, vehicles. When these rules are executed by vehicles, a global pattern emerges. The current basic set of rules that are used to carry out **Join** and **Exit** maneuvers are laid out below and will be described in the context of the maneuvers that use them.

- **R1:** $0 \leq PAP_{gap} < TPP^{L}_{gap} \rightarrow$ Accelerate
- **R2:** $TPP^{L}_{gap} \leq PAP_{gap} \leq TPP^{U}_{gap} \rightarrow$ Maintain Speed
- **R3:** $SAP_{gap} == 0 \rightarrow$ Become Tail Member
- **R4:** $PAP_{gap} > TPP^{U}_{gap} \rightarrow$ Decelerate
- **R5:** $PAP_{gap} == 0 \rightarrow$ Change controller to ACC

b) **Join Maneuver:** We currently assume that vehicles join the platoon at the tail and both the `JOINER` and the platoon are in the same lane. In our future work, these assumptions will be relaxed by implementing the *Alignment* rule to allow the `JOINER` to change lane and by enabling vehicles to join anywhere in the platoon. Vehicles are by default in the `TAIL-MEMBER` role and self-assign a globally unique platoon-id, e.g., a UUID [22]. When a vehicle enters a freeway, it uses the ACC mode and scouts for platoons ahead. When it senses PP_{gap} from a `TAIL-MEMBER` of another platoon, it resets its PAP_{gap} to zero and changes its role to `JOINER`. Using rule **R1**, the `JOINER` catches up and joins the platoon by communicating with the platoon's current `TAIL-MEMBER` as depicted in Figure 2. After completing the **Join** maneuver, the `JOINER` changes its role to the `TAIL-MEMBER` and adopts the platoon-id of the platoon, replacing its original platoon-id. The previous `TAIL-MEMBER` changes its role to `IN-MEMBER.` All vehicles, except the platoon's first member, maintain their speed using rule **R2**. Note that rules **R1, R4** are used to adjust to the changes in the first vehicle's speed.

TABLE II: Simulation Parameters

	Parameter	Value
comm	TX pwr (centralized)	20 dBm
	TX pwr (emergent)	10 dBm
	Epoch (δt)	0.1s
mobility	Platoon size	20 vehicles
	Desired gap	15m
	$MAXBOFF$	3s
	h	0.5s
	L_i	4m
	r	2m
	k_p, k_d, k_{dd}	0.2, 0.7, 0
	τ	0.5s

c) **Exit Maneuver:** Any vehicle can leave the platoon without the need to take specific action (e.g., inform other vehicles). When exiting the platoon, the vehicle generates a new globally unique platoon-id which will be broadcasted as part of vehicles' new pheromones (i.e., once it leaves the platoon). Pheromones received with different platoon-id's are dropped by the vehicles. When the `TAIL-MEMBER` (V_n) leaves the platoon, V_{n-1} will not receive any SP_{gap} from its successor. Therefore, SAP_{gap} nears zero and V_{n-1} becomes the `TAIL-MEMBER` using rule **R3**. When an `IN-MEMBER` (V_i) leaves the platoon, V_{i+1} catches up with V_{i-1} using rule **R1**. Lastly, when the first member V_1 leaves the platoon, V_2 will no longer receive periodic PP_{gap} from V_1 and PAP_{gap} will become less than TPP^{L}_{gap}; therefore, V_2 will accelerate to catch-up due to rule **R1**. However, PAP_{gap} soon will near zero and V_2 will become the platoon's first member and, using rule **R5**, will change its mode to ACC from CACC, maintaining its speed independently without cooperation from any other vehicle.

V. Evaluation

A. Experimental Methodology

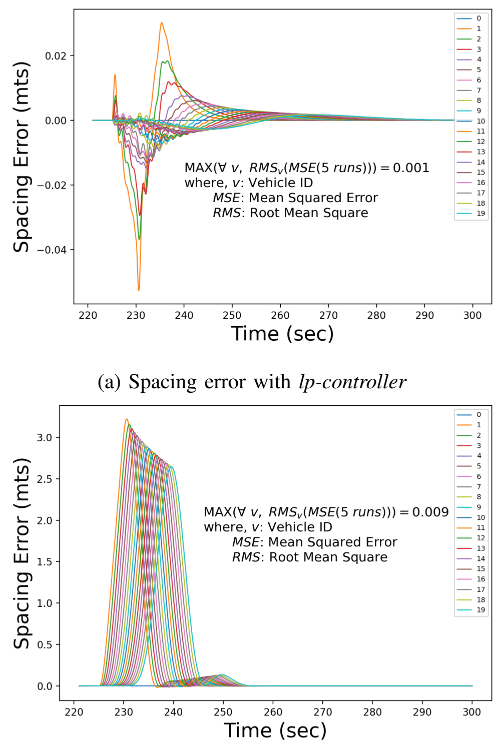
We evaluate our novel emergent platooning system by comparing the performance of its controller which is based on the *ploeg-controller*, as well as its **Join** and **Exit** maneuvers against centralized platooning which uses the (*lp-conroller*) [20]. We conducted our experiments using the PLEXE 2.1 simulator [9], which in turn uses SUMO [23] for simulating vehicular traffic, and OMNeT++ [24] for simulating the communication network. We implemented our emergent platooning on

PLEXE 2.1 and extended its default centralized platooning implementation to include the **Exit** maneuver. Our evaluation uses PLEXE 2.1's default simulation parameters [9]. Table II summarizes our simulation parameters. Due to space limitations, we only list new parameters as well as parameters whose values are different from PLEXE's 2.1 default parameters.

a) **Performance Metrics:** We use *spacing error* to evaluate controller performance and maneuver latency, i.e., the time a vehicle stays in the maneuvering state, to evaluate the performance of our emergent platooning approach. For **Joins**, maneuver latency is given by the total time from when a joining vehicle enters the simulation till it becomes a part of the platoon and for **Exits**, it is the total time from when a vehicle decides to exit the platoon till it disassociates itself from the platoon.

In our current implementation, we handle message loss by having the `JOINER` and `TAIL-MEMBER` abort the maneuver if they do not hear from one another within a specified time interval and retry the maneuver subsequently. We will consider more advanced reliability mechanisms in the future.

b) **Experiments:** Once a simulation run starts, every $2s$ a new vehicle enters the scene until a total of twenty vehicles are present. All vehicles enter the system in ACC mode and move to CACC mode after joining the platoon, except for the first vehicle of the platoon which continues to be in ACC mode. In emergent platooning, vehicles will start as a `TAIL-MEMBER`. In centralized platooning, the first vehicle in the experiment assumes the role of the leader and coordinates the maneuvering of all the remaining vehicles that join the platoon one after the other. The vehicle closest to the leader in case of centralized platooning and the one closest to the `TAIL-MEMBER` in case of emergent platooning will be allowed to join. Proximity is determined by vehicle's position information contained in the `JOIN-REQ`. Using negative `JOIN-RESP` messages, the leader or the `TAIL-MEMBER` coordinating the maneuver will reject requests from the vehicles that are not close enough or if they are coordinating another maneuver. To simulate worst case **Exit** behavior, all but the first and the tail vehicles decide to exit the platoon simultaneously, once all the vehicles have joined the platoon and the platoon has remained in steady state for a few seconds. All experiments were run 5 times with different random seeds and results reported in Sections V-B and V-C are averages over the 5 runs including maximum mean error (Figure 3) and standard deviation (Figure 5).



(a) Spacing error with *lp-controller*

(b) Spacing error with *ploeg-controller*

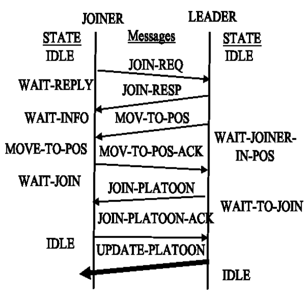
Fig. 3: Spacing Error

B. Controller Performance

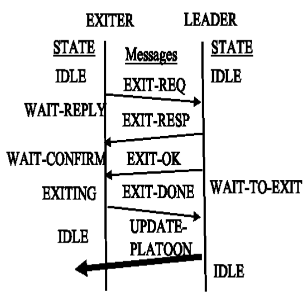
We compare the performance of the *ploeg-controller* used in our emergent platooning against *lp-controller* used in centralized platooning. The resulting spacing error is evaluated after causing a "disturbance" in the system. In our experiments, after the platoon is formed, a disturbance is introduced at time $t = 225s$ by having the leader accelerate to $125km/h$ and then decelerate back to its original speed of $100km/h$ after $5s$.

The headway (h) for the *ploeg-controller*, which uses constant-time spacing, is set to $0.5s$, the minimum necessary headway for string stability [21] given that gap-pheromone messages are transmitted every 100ms (δt). The equivalent desired gap for the *lp-controller*, which uses constant-distance spacing policy, is ($15m$), which is what we use in our experiments. Figure 3 shows the spacing error (in meters) experienced by the different vehicles in the platoon over time after the disturbance is introduced, where a positive or a negative value means that the spacing is higher or lower than the desired spacing, respectively. Figure 3 indicate that the spacing error decreases towards the tail of the platoon for both controllers, which demonstrates that the platoon is string stable. Note that in the case of *ploeg-controller*, the vehicles towards the tail end of the platoon show a maximum gap of ≈3m, almost 100x higher than the spacing errors in the case of *lp-controller*. This is because, in our emergent platooning system, each vehicle relies only on the information from its predecessor, whereas under the *lp-controller*, all vehicles have access to the leader's information that enables faster convergence. It is important to note that the spacing does not fall below 0.05m and 0.01m from the desired spacing in the *lp-controller* and *ploeg-controller*, respectively. This means that the vehicles were never at the risk of colliding. However, it is interesting to note that the desired spacing is reached almost ≈3x faster after disturbance is introduced in the *ploeg-controller* (≈30s), compared to the *lp-controller* (≈80s). As part of future work, we will extend the *ploeg-controller* to use information from two predecessors which have been shown to result in smaller headway [25].

C. Join Maneuver Performance



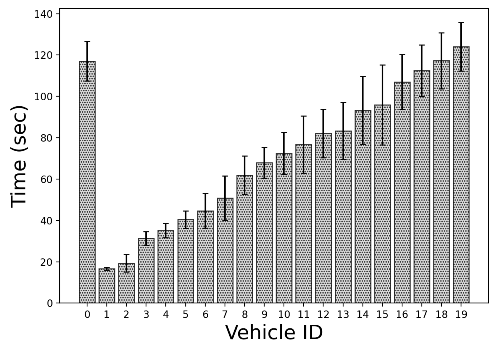
(a) Join



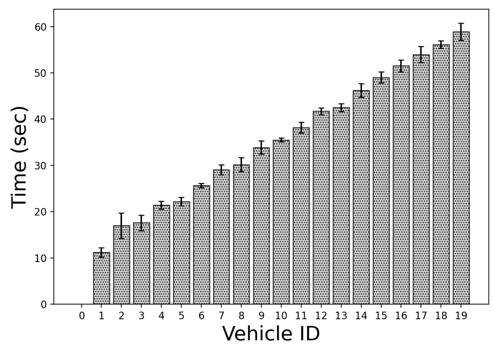
(b) Exit

Fig. 4: Centralized maneuver message exchange

Figure 4a shows the message exchange during **Join** maneuvers in the centralized approach. Joining vehicles enter the maneuvering state as soon as they enter the system and remain in that state until they are let into the platoon by the leader. As such, we observe in Figure 5a that vehicle 19, which is the last to enter the experiment and thus, the platoon, spends the longest time ($124.03s$) in **Join** maneuvering state. The leader, vehicle 0, will be coordinating only one vehicle at a time and join requests from any other vehicle will be rejected. Vehicles requesting to join the platoon backoff for a random interval ($0 < backoff < MAXBOFF$) before retrying. While lower $MAXBOFF$ leads to higher retries, $MAXBOFF$ that are too long result in higher maneuvering latency. In our experiments, $MAXBOFF = 3s$. The leader remained in maneuvering state for a total of $117.03s$. When we increased $MAXBOFF$, the time the leader remained in maneuvering state decreased.



(a) Centralized platooning



(b) Emergent platooning

Fig. 5: Average time to complete the **Join** maneuver

For example, for $MAXBOFF = 20s$, vehicle 19 experienced higher **Join** maneuvering latency ($216.45s$), but the leader maneuvering time decreased to $62.42s$. This is because `JOINERS` waited longer to retry joining after their requests were rejected.

In emergent platooning, each vehicle (except the first and the tail) will be in the **Join** maneuvering state twice, once as `JOINER` and later as `TAIL-MEMBER`, which coordinates the maneuver. The average amount of time the `TAIL-MEMBER` spent **coordinating** the maneuver is just $3.8ms$ compared to $117.03s$ spent by the leader in the centralized approach. This discrepancy is due to the fact that in centralized platooning, the leader has to coordinate all maneuvers, while in emergent platooning the coordination is localized between the current `TAIL-MEMBER` and the `JOINER`. According to Figure 5b, the time to complete all **Joins** in the emergent system is $58.85s$, which represents a 50% reduction compared to centralized platooning. This reduction is due to the absence of serialization and the backoff during the maneuver. Backoff was not required in emergent platooning as the vehicles use lower power for communicating with immediate neighbors compared to centralized platooning that leads to lower signal interference.

D. Exit Maneuver Performance

Prior work has focused mostly on single vehicle exits. However, since in real-world scenarios, multiple vehicles can exit a platoon simultaneously, our implementation is able to handle both single- and multi-vehicle exits. The message exchange between the exiter and the leader in centralized platooning is shown in Figure 4b. The leader has to serially coordinate the exit of each vehicle. Therefore, the average time each vehicle has to wait before it exits the platoon is higher. In our experiments, the tail vehicle had to wait for approximately $2.5s$, while the leader was in maneuvering state for a few milliseconds for each vehicle. In emergent platooning, the exiting vehicle need not inform any vehicle and consequently, there are no messages exchanged and exits are "instantaneous". The tail vehicle's pheromone concentration reduces due to the absence of its predecessor, which causes it to accelerate to catch-up until it senses pheromone from the first vehicle after all vehicles exit. At that point, it coordinates with the first vehicle to complete platoon consolidation.

VI. Conclusion and Future Work

In this paper, we presented a novel emergent-behavior based decentralized platooning framework. Our preliminary complexity and latency results for the **Join** and **Exit** maneuvers indicate that our platooning approach is a promising alternative to centralized platooning. We also demonstrated experimentally that our emergent platooning achieves string-stability after disturbance; while emergent platooning is able to achieve the desired inter-vehicular spacing faster (e.g., after a disturbance), its spacing error is higher when compared to centralized platooning. As part of future work, we plan to improve emergent platooning's spacing error caused by disturbances as well as implement additional maneuvers.

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